The recursion equations for ordinal addition

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openlogic:set-theory/ord-arithmetic:local-snapshot-20261013; CC BY 4.0

Editorial scope. Statement and complete author proof are included verbatim as the span lines 477-507 of the pinned original source; no text is rewritten or paraphrased.

Allowed standard prerequisites (editorial).

- ordinal addition defined via the lexicographic order type of a two-piece concatenation
- strict suprema of sets of ordinals
- initial segments of ordinal order types

Lemma 0.1. *For any ordinals α, β, we have:*

$$\begin{aligned} \alpha + 0 &= \alpha \\ \alpha + (\beta + 1) &= (\alpha + \beta) + 1 \\ \alpha + \beta &= \subsetneq_{\delta<\beta} (\alpha + \delta) && \text{if } \beta \text{ is a limit ordinal} \end{aligned}$$

Proof. We check case-by-case; first:

$$\begin{aligned} \alpha + 0 &= \mathrm{ord}((\alpha \times \{0\}) \cup (0 \times \{1\}), \prec) \\ &= \mathrm{ord}((\alpha \times \{0\}) \cup \{0\}, \prec) \\ &= \alpha \\ \alpha + (\beta + 1) &= \mathrm{ord}((\alpha \times \{0\}) \cup (+1\beta \times \{1\}), \prec) \\ &= \mathrm{ord}((\alpha \times \{0\}) \cup (\beta \times \{1\}), \prec) + 1 \\ &= (\alpha + \beta) + 1 \end{aligned}$$

Now let $\beta \neq \emptyset$ be a limit. If $\delta < \beta$ then also $\delta + 1 < \beta$, so $\alpha + \delta$ is a proper initial segment of $\alpha + \beta$. So $\alpha + \beta$ is a strict upper bound on $X = \{ \alpha + \delta \mid \delta < \beta \}$. Moreover, if $\alpha \leq \gamma < \alpha + \beta$, then clearly $\gamma = \alpha + \delta$ for some $\delta < \beta$. So $\alpha + \beta = \subsetneq_{\delta<\beta} (\alpha + \delta)$. □

Proof. We check case-by-case; first:

$$\begin{aligned}
\alpha + 0 &= \mathrm{ord}((\alpha \times \{0\}) \cup (0 \times \{1\}), \prec) \\
&= \mathrm{ord}((\alpha \times \{0\}) \cup \{0\}, \prec) \\
&= \alpha \\
\alpha + (\beta + 1) &= \mathrm{ord}((\alpha \times \{0\}) \cup (+1\beta \times \{1\}), \prec) \\
&= \mathrm{ord}((\alpha \times \{0\}) \cup (\beta \times \{1\}), \prec) + 1 \\
&= (\alpha + \beta) + 1
\end{aligned}$$

Now let $\beta \neq \emptyset$ be a limit. If $\delta < \beta$ then also $\delta + 1 < \beta$, so $\alpha + \delta$ is a proper initial segment of $\alpha + \beta$. So $\alpha + \beta$ is a strict upper bound on $X = \{\, \alpha + \delta \mid \delta < \beta \,\}$. Moreover, if $\alpha \leq \gamma < \alpha + \beta$, then clearly $\gamma = \alpha + \delta$ for some $\delta < \beta$. So $\alpha + \beta = \subsetneq_{\delta < \beta} (\alpha + \delta)$. □